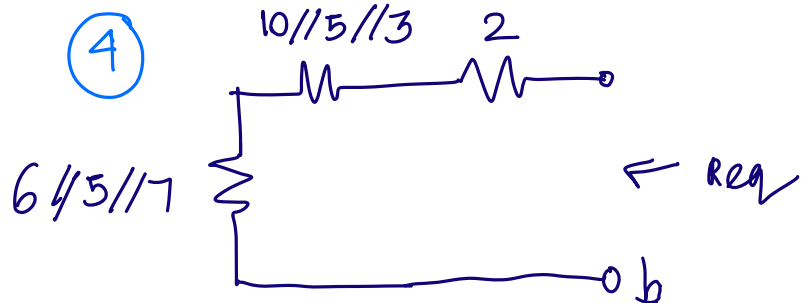
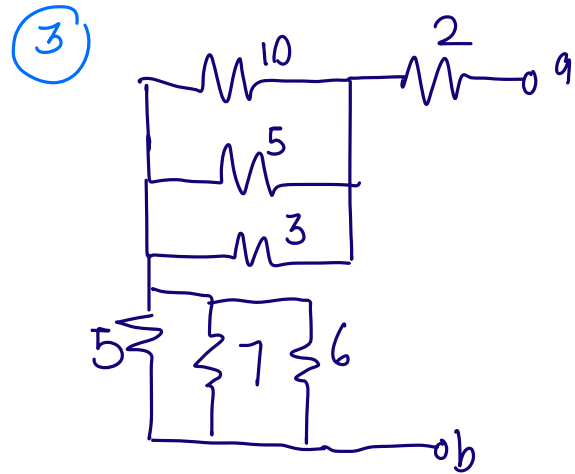
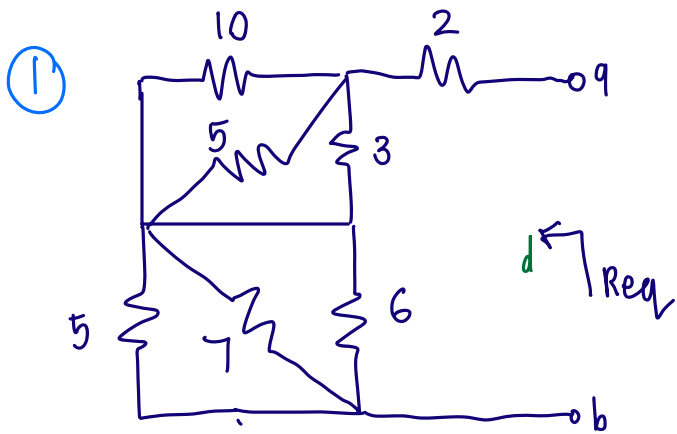
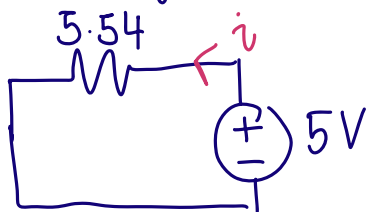


Q1 (a) (i) Req?



$$\begin{aligned}
 R_{eq} &= (10//5//3) + 2 + (5//7//6) \\
 &= 30/9 + 2 + 210/107 \\
 &= 5.54 \Omega
 \end{aligned}$$

(ii) 5V battery to a-b



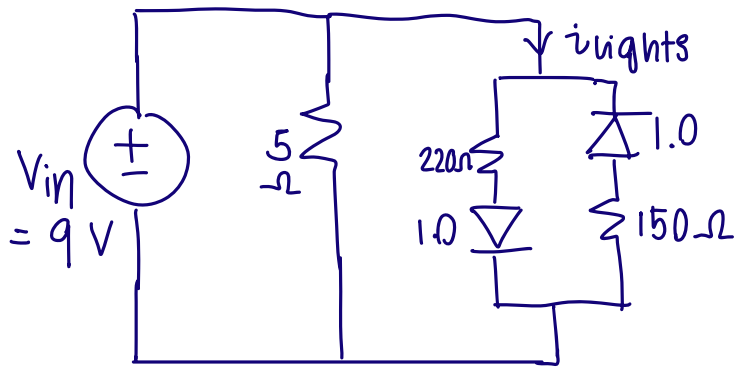
$$\begin{aligned}
 i &= V/R \\
 &= 5/5.54 \\
 &= 0.903 \text{ A}
 \end{aligned}$$

(iii)

$$\begin{aligned}
 P &= i^2 R_{eq} \\
 &= 0.903^2 (5.54) \\
 &= 4.517
 \end{aligned}$$

$$\begin{aligned}
 P &= \frac{\Delta W}{\Delta t} \Rightarrow \Delta W = P \Delta t \\
 &= 4.517 \times 2 \times 3600 \text{ s} \\
 &= 32.525 \text{ kJ}
 \end{aligned}$$

Q1(b)(i) i_{lights} when RED is lit



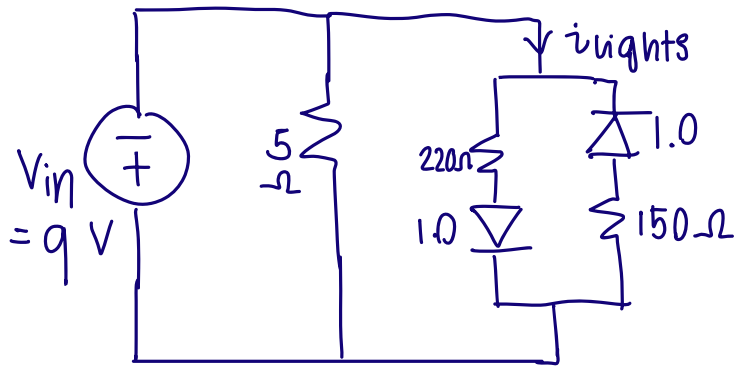
$$i_{lights} = \frac{V_{in} - V_{red}}{220}$$

$$= \frac{9 - 1}{220}$$

$$= 0.036 \text{ A}$$

unavailable

(ii) i_{lights} when GREEN is lit



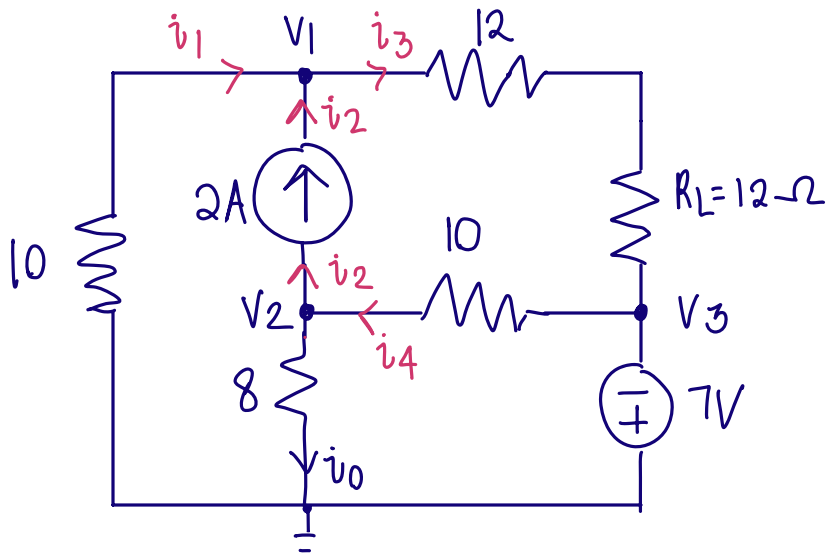
$$i_{lights} = - \frac{(V_{in} - V_{green})}{150}$$

$$= - \left(\frac{9 - 1}{150} \right)$$

$$= -0.053 \text{ A}$$

available

Q2 (a) (i) I_0 & V_2 ?



KCL at node V_1 :

Redundant KCL (for part a)) used in (ii)

$$\begin{aligned} i_1 + i_2 &= i_3 \\ -\frac{V_1}{10} + 2 &= \frac{V_1 - V_3}{12 + 12} \\ -\frac{V_1}{10} + 2 &= \frac{V_1}{24} - \frac{V_3}{24} \\ -\frac{17}{120} V_1 + \frac{V_3}{24} &= -2 \quad (1) \end{aligned}$$

$$V_3 = -7 \quad (3)$$

(3) into (2) :

$$\begin{aligned} \frac{9V_2}{40} - \frac{V_3}{10} &= -2 \\ \frac{9V_2}{40} - \frac{(-7)}{10} &= -2 \end{aligned}$$

$$V_2 = -12V$$

$$i_0 = -12/8$$

$$= -1.5A$$

KCL at node V_2 :

$$\begin{aligned} i_2 + i_0 &= i_4 \\ 2 + \frac{V_2}{8} &= \frac{V_3 - V_2}{10} \\ \frac{9V_2}{40} - \frac{V_3}{10} &= -2 \quad (2) \end{aligned}$$

KCL V_1 :

$$(ii) \quad -\frac{17}{120} V_1 - \frac{7}{24} = -2$$

$$V_1 = 12.06V$$

voltage divider
↓

$$V_{RL} = (V_1 - V_3) \left(\frac{12}{24} \right)$$

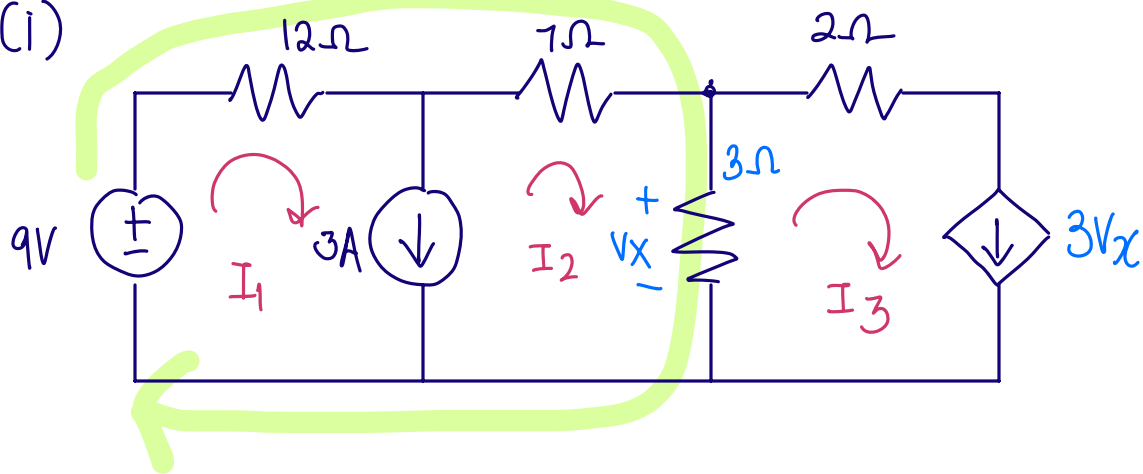
$$= (12.06 - 7) \left(\frac{12}{24} \right)$$

$$= 9.53V$$

$$P_{RL} = \frac{9.53^2}{12}$$

$$= 7.57W$$

(b) (i)



Constraint Eqn:

$$V_x = 3(I_2 - I_3) \\ = 3I_2 - 3I_3$$

$$I_1 - I_2 = 3 \quad \text{--- (2)}$$

$$I_3 = 3V_x \\ V_x = I_3/3$$

$$3I_2 - 3I_3 = I_3/3 \\ 9I_2 - 9I_3 = I_3 \\ 9I_2 - 10I_3 = 0 \quad \text{--- (3)}$$

Mesh

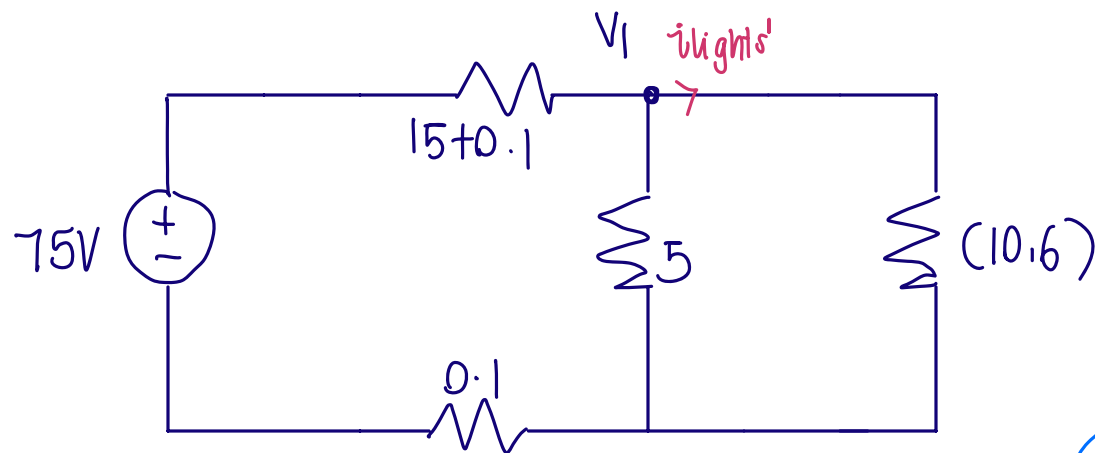
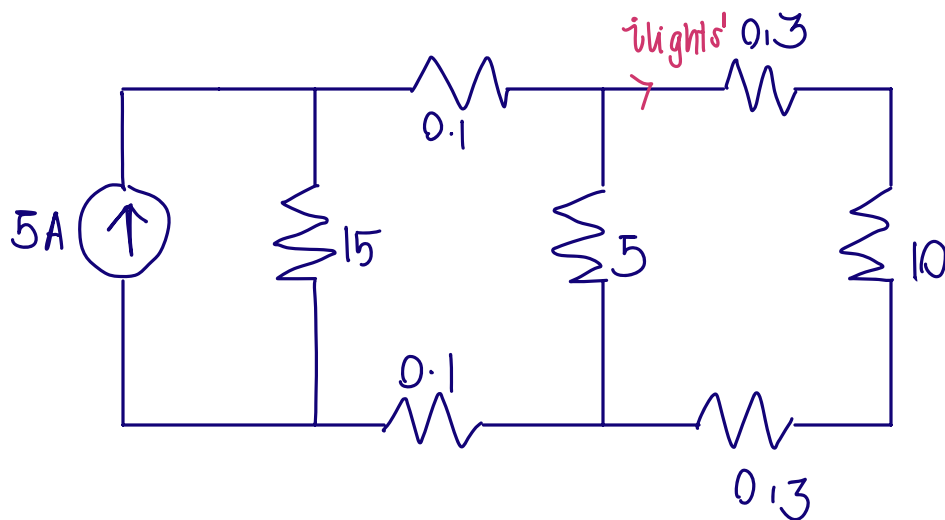
$$\begin{aligned} -9 + 12(I_1) + 7(I_2) + 3(I_2 - I_3) &= 0 \\ 12I_1 + 7I_2 + 3I_2 - 3I_3 &= 9 \\ 12I_1 + 10I_2 - 3I_3 &= 9 \quad \text{--- (3)} \end{aligned}$$

EQUATIONS:

$$\begin{aligned} 9I_2 - 10I_3 &= 0 \\ I_1 - I_2 &= 3 \\ 12I_1 + 10I_2 - 9I_3 &= 9 \end{aligned}$$

Q3 (i) i_{lights} ?

• Contribution from current source: (Voltage source short circuit)



source transformation

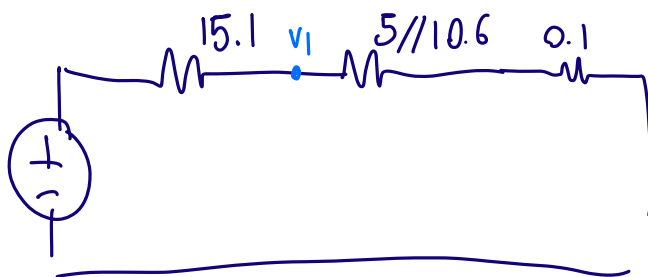
$$V = IR \\ = 5(15) \\ = 75V$$



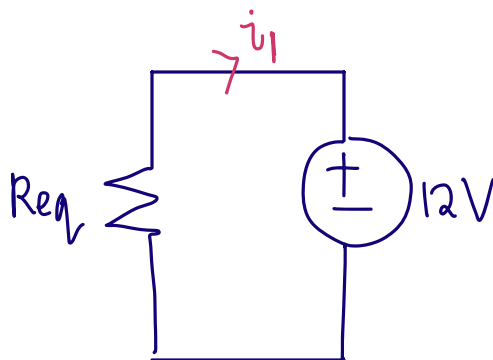
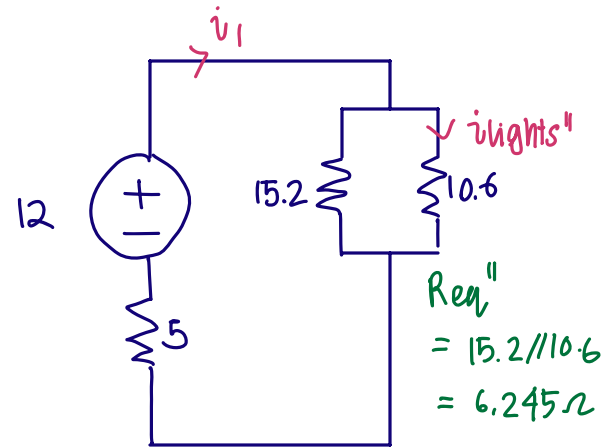
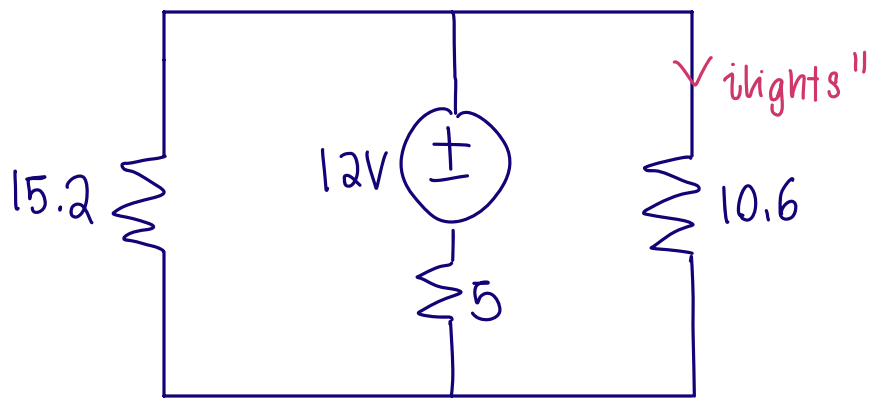
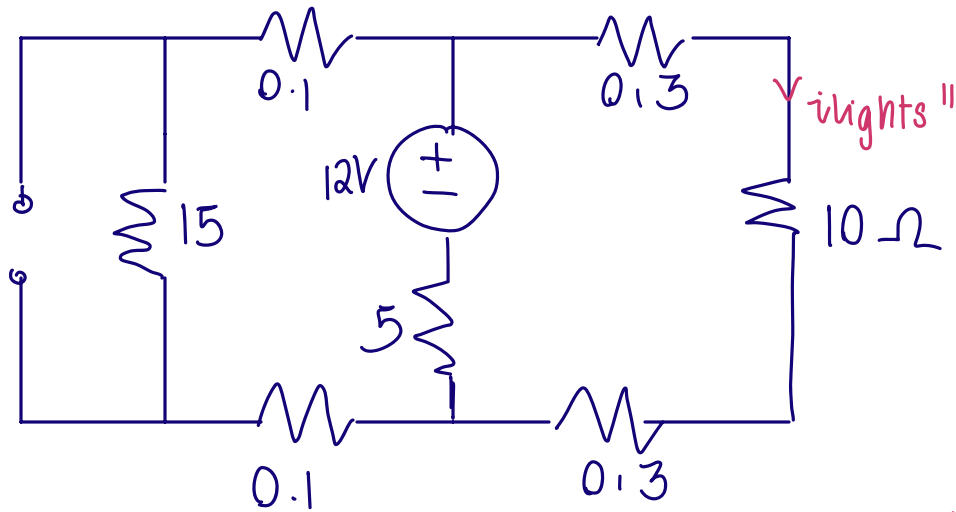
$$V_1 = \frac{(5 // 10.6)}{(5 // 10.6) + 15.1 + 0.1} (75)$$

$$= 13.701V$$

$$i_{lights'} = \frac{13.701}{10.6} \\ = 1.293A$$



- Contribution of the voltage source, current source open circuit.



$$R_{eq} = 15.2 // 10.6 + 5$$

$$= 11.245\Omega$$

$$i_1 = 12 / R_{eq}$$

$$= 1.067A$$

$$i_{lights}'' = 1.067 \left(\frac{6.245}{10.6} \right)$$

$$= 0.629A$$

$$i_{lights} = i_{lights}' + i_{lights}''$$

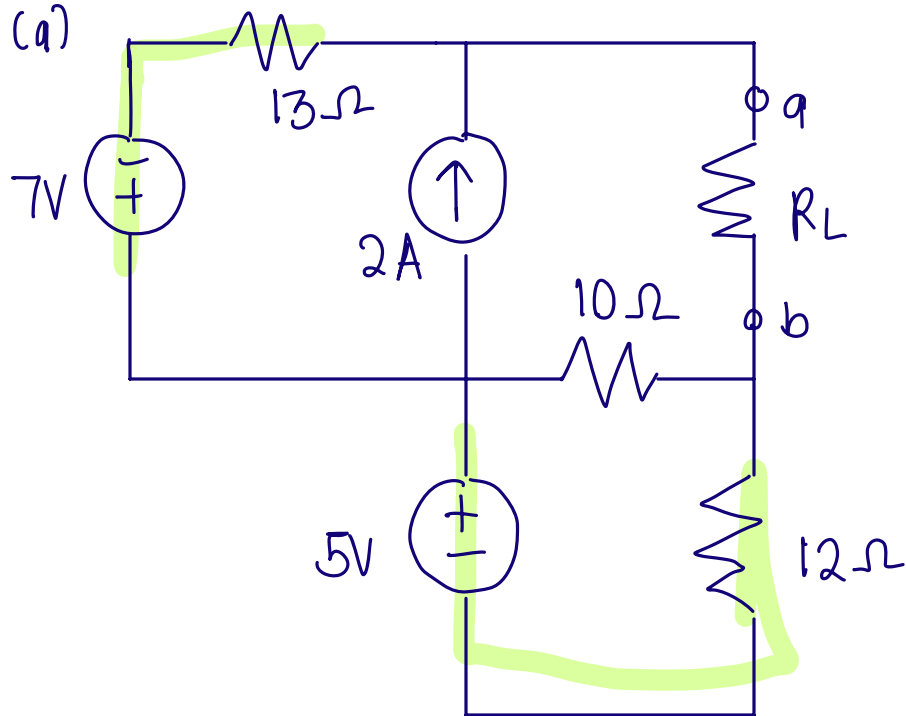
$$= 1.922$$

Gi) $P_{lights} = i^2(R)$

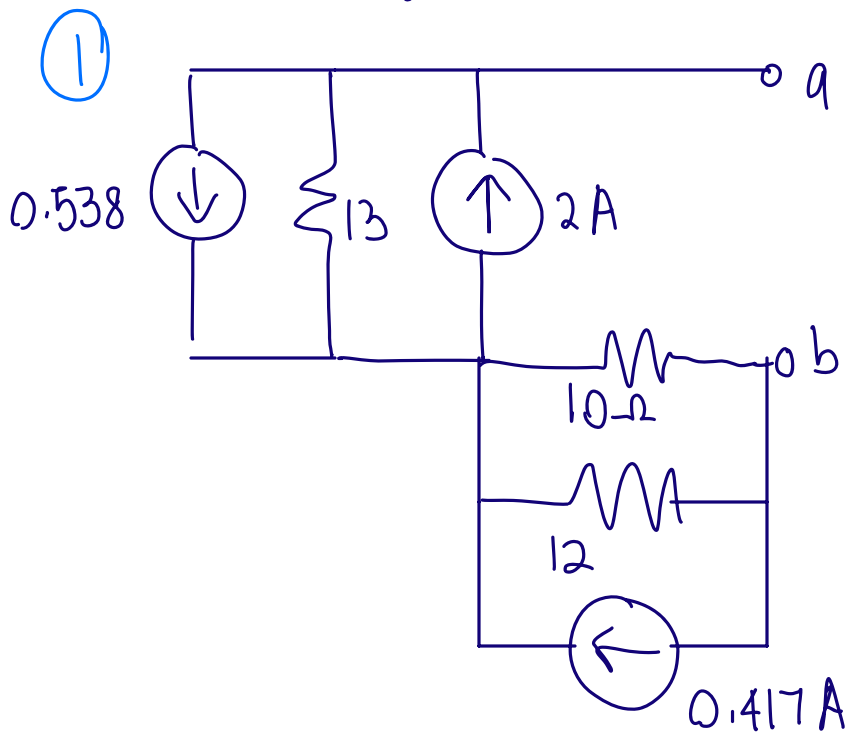
$$= (1.922^2)(10)$$

$$= 36.9W$$

Q4 (a)



Thévenin using source transformation



①

$$V = IR$$

$$7 = 13(I_1)$$

$$I_1 = 7/13$$

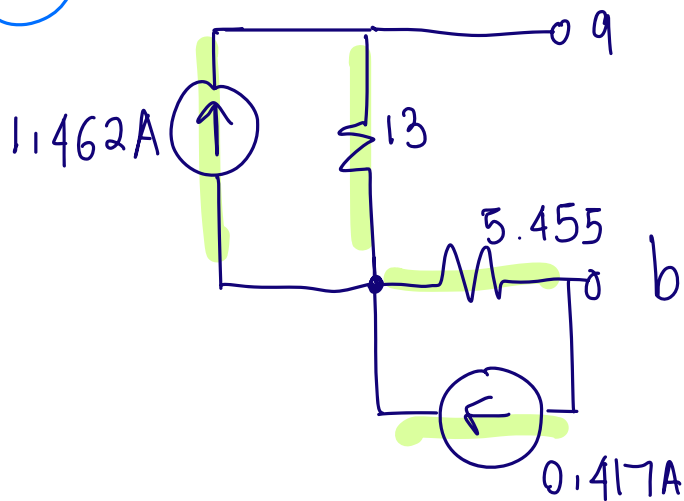
$$= 0.538 \text{ A}$$

$$V_5 = IR$$

$$I_5 = 5/12$$

$$= 0.417 \text{ A}$$

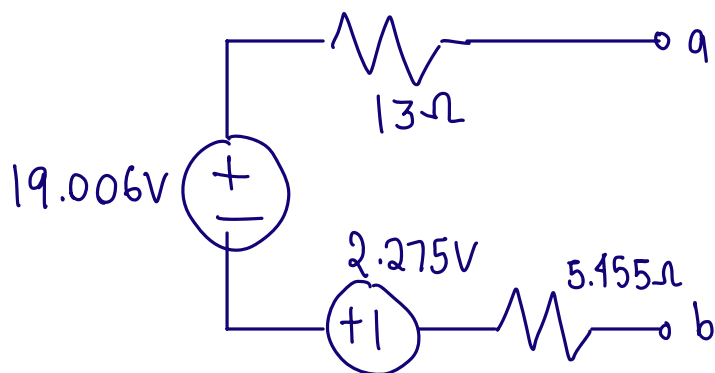
② Combine parallel current sources, parallel resistors



$$I_{\text{new}} = 2 - 0.538 \\ = 1.462 \text{ A}$$

$$10 // 12 = 5.455 \Omega$$

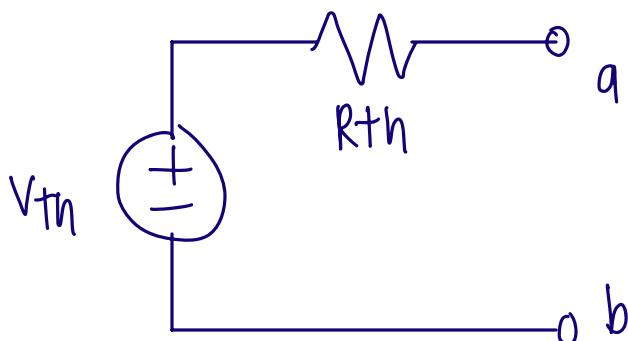
③ change to voltage sources



$$V_1 = 1.462(13) \\ = 19.006 \text{ V}$$

$$V_2 = 5.455(0.417) \\ = 2.275 \text{ V}$$

④ combine everything

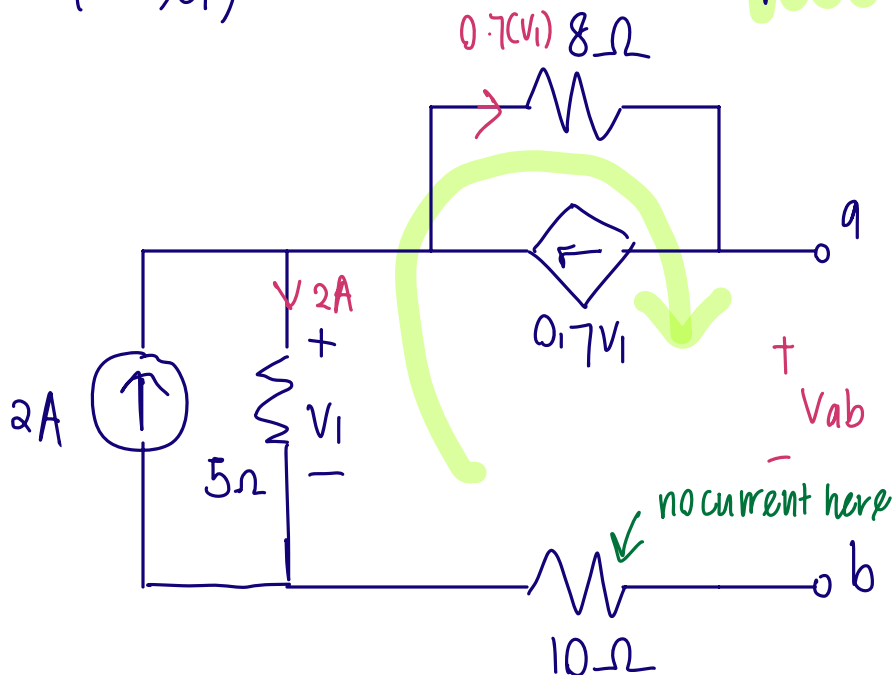


$$R_{th} = 13 + 5.455 \\ = 18.455 \Omega$$

$$V_{th} = 19.006 + 2.275 \\ = 21.281 \text{ V}$$

Q4 (b) Find V_{th}

open-circuit voltage



$$V_1 = 2(5) = 10V$$

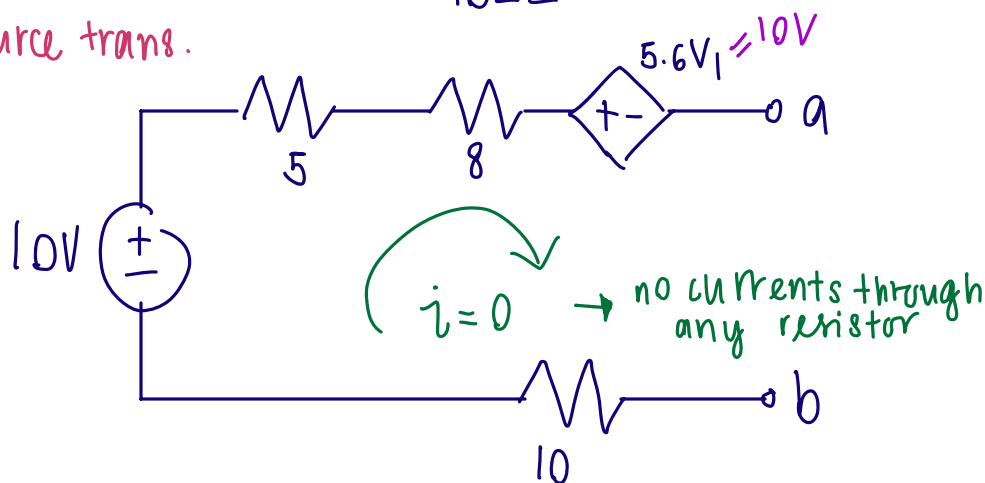
KVL: source transformation

$$-V_1 + 8(0.7V_1) + V_{ab} = 0$$

$$-10 + 8(0.7 \times 10) + V_{ab} = 0$$

$$V_{ab} = -46V$$

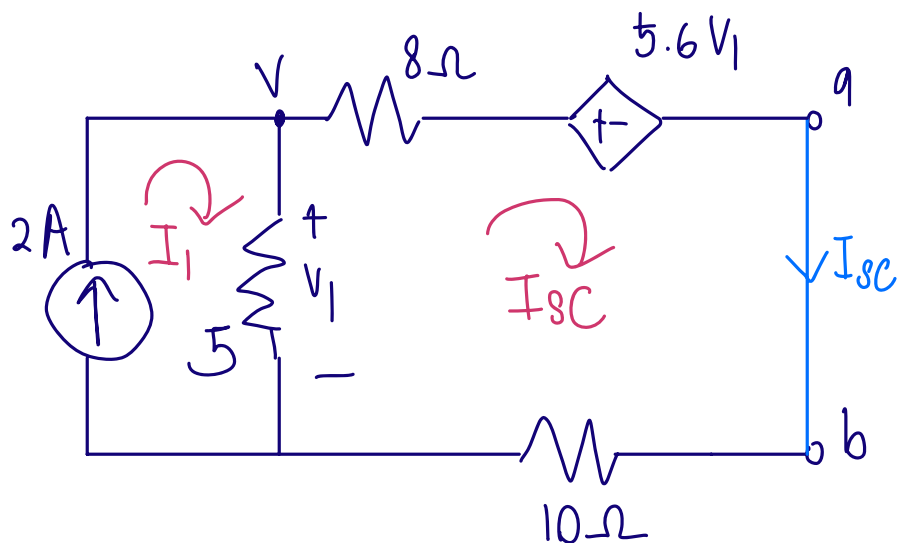
source trans.



$$-10 + 5.6(10) + V_{ab} = 0$$

$$V_{ab} = -46V$$

If used I_{sc} , need R_{th} :



$$I_1 = 2A$$

$$V_1 = 5(I_1 - I_{sc}) = 10 - 5I_{sc}$$

KVL:

$$-5(I_1 - I_{sc}) + 8I_{sc}$$

$$+ 5.6(V_1) + 10I_{sc}$$

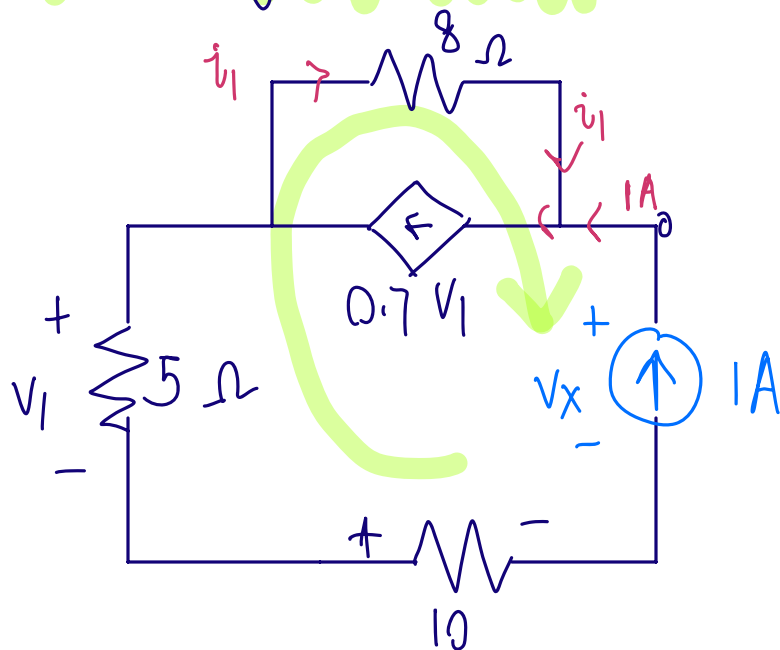
$$= 23I_{sc} - 5I_1 + 56 - 28I_{sc}$$

$$-5I_{sc} + 46 = 0$$

$$I_{sc} = 9.2A$$

(ii) Find $R_{th} = ?$

R_{th} using Test source



$$V_1 + 1 = 0.7 V_1 \Rightarrow 0.3 V_1 = 1 \Rightarrow V_1 = 10/3 \approx 3.33V$$

$$i_1 = 2.5 A$$

$$V_1 = 5(1) = 5V$$

KVL outer loop:

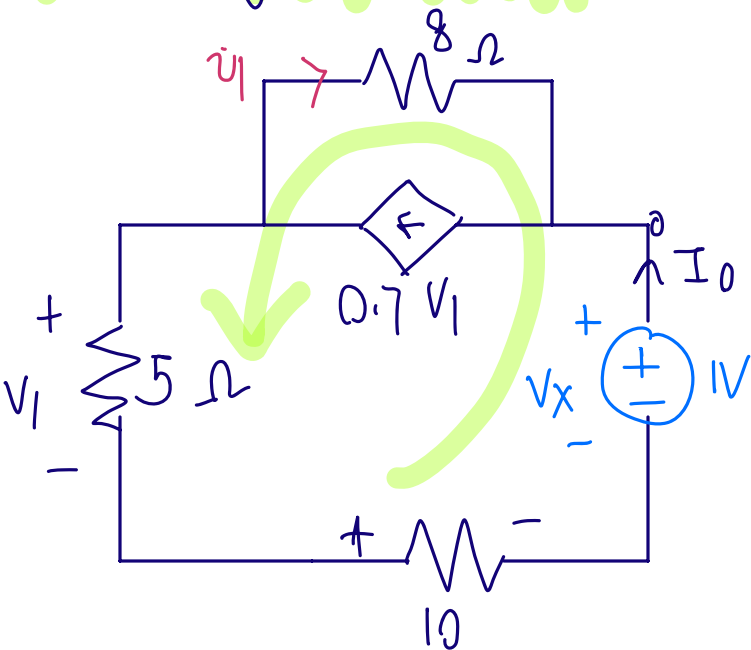
$$-5(1) + 8(i_1) + V_x + (10)(1) = 0$$

$$-5 + 8(2.5) + V_x - 10 = 0$$

$$V_x = -5$$

$$R_{th} = -5 \Omega$$

R_{th} using Test source



$$V_1 = I_0(5)$$

$$V_1 + I_0 = 0.7 V_1$$

$$= 3.5 I_0$$

$$V_1 = 2.5 I_0$$

KVL outer loop:

$$-1 - 8(i_1) + 5 I_0 + 10 I_0 = 0$$

$$-20 I_0 + 5 I_0 + 10 I_0 = 1$$

$$-5 I_0 = 1$$

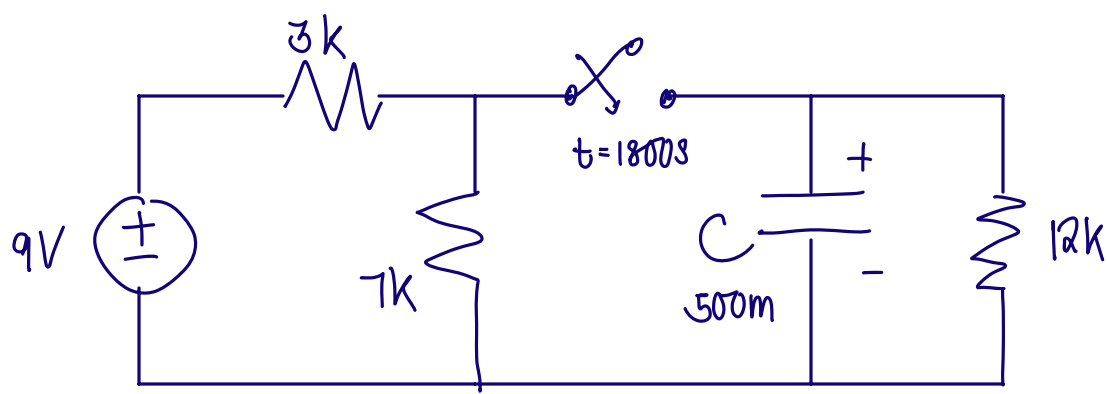
$$I_0 = -1/5$$

$$R_{th} = -5 \Omega$$

cii)

$$\begin{aligned}P_{\max} &= \frac{V+V^2}{4R+h} \\&= \frac{46^2}{4(-5)} \\&= -105.8 \text{ W}\end{aligned}$$

Q5



(i) $V_o(t)$, $0 < t \leq 1800s$

Step response:

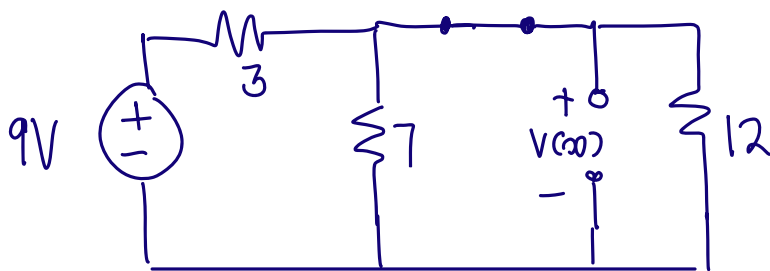
$$V(t) = V(\infty) + [V(t_0) - V(\infty)] e^{-\frac{(t-t_0)}{\tau}} , \tau = RC$$

→ When switch is open :

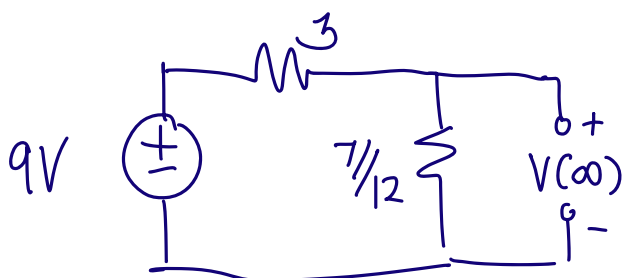
- $V(0) = 0$ for $t < 0$

→ when switch is closed :

- $V(\infty) = ?$



$$7 // 12 \rightarrow 4.421 \Omega$$

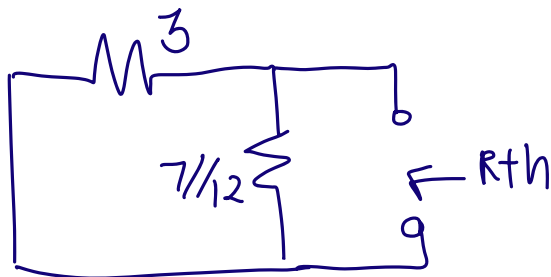


Voltage divider:

$$V(\infty) = (9) \left(\frac{4.421}{4.421 + 3} \right)$$

$$= 5.362 V$$

• calculate $\tau = R_{th} C$



$$R_{th} = 3 \parallel (4.421) \\ = 1.787 k\Omega$$

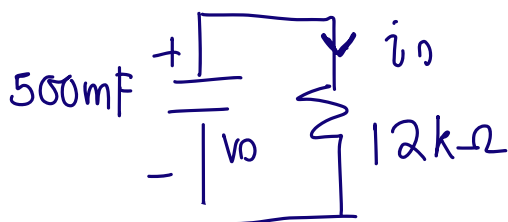
$$\tau = R_{th} C \\ = 1.787 k(500m) \\ = 893.5$$

Substitute:

$$V_0 = 5.362 + [0 - 5.362] e^{-t/893.5} \\ = 5.362 [1 - e^{-t/893.5}] V$$

(ii) V_0 , when $t = 1800s$

↳ switch open:



Natural response

$$V_0 = V(1800) e^{-\left(\frac{t-1800}{\tau}\right)}, \tau = RC$$

$$V(1800s) = 5.362 (1 - e^{-\frac{1800}{893.5}}) \\ = 4.645 V$$

$$V_0(t) = 4.645 e^{-\left(\frac{t-1800}{6000}\right)} V$$

(iii) $i_0(t)$, $t > 1800$:

$$V_0 = 4.645 e^{-\left(\frac{t-1800}{6000}\right)}$$

$$i_0 = C \frac{dV_0}{dt} = 0.5 \frac{d}{dt} (4.645 e^{-\left(\frac{t-1800}{6000}\right)}) \\ = 0.5 (4.645) \left(-\frac{1}{6000}\right) e^{-\left(\frac{t-1800}{6000}\right)} \\ i_0 = -0.000387 e^{-\left(\frac{t-1800}{6000}\right)} A$$

$$\tau = RC \\ = 12 k(500m) \\ = 6000s$$